MISP Deployment

Some basic guidelines

CIRCL / Team MISP Project

MISP Project

CRI MISP 2024

MISP

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MISP deployment considerations

- **Deployment types**
- **Distro** choice
- **Hardware specs**
- **Authentication**
- Other considerations - **settings**, **gotchas**

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- Distro choice
- Hardware specs
- Authentication
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Deployment types

- Native install
  - Manual
  - One-liner script - `INSTALL.sh`: `https://github.com/MISP/MISP/tree/2.4/INSTALL`
- MISP VM: `https://www.circl.lu/misp-images/latest/`
- Docker: `https://github.com/MISP/misp-docker`
- RPM maintained by `amuehlem`: `https://github.com/misp/misp-rpm`
- Cloud provider images: `https://github.com/MISP/misp-cloud`
- LXC images via `misp-airgap`: `https://github.com/MISP/misp-airgap/`

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Docker options

- There are several MISP Docker installations available.
- The official MISP Docker installation is maintained by `ostefano` and can be found at `https://github.com/MISP/misp-docker`.
- The official MISP Docker image also includes one for the MISP modules.

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Distro options

- Ubuntu 22.04 (20.04 will also work)
  - Our target platform
  - Our CI target
  - Use this unless you are absolutely forced not to
  - This is the platform we can support you with!
- CentOS 7
  - Annoying to operate
  - Less tested, though used by many
  - CentOS is dead. Consider other options
- RHEL 7
  - Same annoyance as CentOS in general
  - We test against CentOS in general, some assembly may be required

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- No firm recommendations, it's highly usage dependent
- It's better to go a bit over what you need than under
- **SSDs** are massively beneficial
- Let's look at what affects specs and some sample configurations

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Hardware considerations

- What are the factors that can impact my performance?
  - Clustering of the data (how many datapoints / event?) (RAM, disk speed)
  - Correlation (RAM, disk speed, disk space)
    - Consider blocking overtly correlating values from doing so
    - Feed ingestion strategy is crucial
  - Over-contextualisation (RAM, disk speed)
    - Tag/attach galaxies to the event instead of each attribute when possible

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  - Logging strategy (Disk space)
  - API users especially with heavy searches (substring searches for example) (RAM, CPU, Disk speed)

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  - Warninglist usage
  - Number of raw attributes on the instance
  - Number of sync connections / recurring syncs (with measure)
  - Tools feeding off the automation channels (ZMQ, kafka, syslog)

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- Username/password is the default
- Some built in modules by 3rd parties (LDAP, Shibboleth, x509, OpenID, Azure Active Directory)
- CustomAuth system for more flexibility
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Other considerations - tuning

- PHP tuning
  - Maximum memory usage (per process)
  - Timeout settings
  - Consider setting it per role!
  - Background processes are exempt
- MySQL: key buffer size is important
- Generally, tune for few heavy requests rather than many light ones

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Other considerations - high availability

- Clustering
  - Load balanced apache servers with MISP
  - Replicating / mirrored database backends
- Careful about session pinning
- Attachment storage can be abstracted / network attached
- An example implementation for AWS
  `https://github.com/0xtf/HAMISPA`

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